

Mahmoud Alyosify

Specialised Machine Learning Engineer | End-to-End ML Pipelines, Training & Deployment | MSc Artificial Intelligence, Queen's University (Canada)

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🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

Machine Learning Engineer who takes models from **raw data to a deployed, monitored service**: feature engineering, training, systematic hyperparameter search, honest evaluation, and packaging for production via **ONNX, Docker, FastAPI and AWS**. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**. Delivered supervised, self-supervised and reinforcement-learning systems — including a **42-experiment** vision sweep reaching **84.30% top-1 accuracy** with ONNX/FAISS retrieval, a satellite collision-risk model operating on a **48-hour** decision window, and a fine-tuned LLM served on AWS. Strong software-engineering base from national-scale backend systems, and **5 years** teaching ML to **10,000+ learners**.

EDUCATION

- MSc in Artificial Intelligence** Sep 2025 – Nov 2026
Expected Nov 2026
Queen's University, Kingston, Ontario, Canada
- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
 - Coursework: **Deep Learning, Large Language Models, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines.**
 - **Master's Research Project (CISC 898)**: training-free decoding-time efficiency for language models — Supervisor: Dr. Rahatara Ferdousi.
- B.Sc. Computer & Information Science (Bioinformatics)** Sep 2019 – Jul 2023
GPA: 3.53 / 4.00
Assiut University, Egypt
- Graduation Project: **Vitalism Solution** — contactless vital-sign estimation via rPPG. **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

ML & Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CNNs, Self-Supervised & Contrastive Learning, Reinforcement Learning (PPO, DQN), Transformers, Fine-Tuning (LoRA, QLoRA)
Modelling Workflow	Feature Engineering, Class-Imbalance Handling, Cross-Validation, Hyperparameter Optimisation (Optuna), Model Selection, Explainability (SHAP), Ablation Studies, Error Analysis
Data Engineering	Pandas, NumPy, PySpark, ETL Pipelines, SQL, Data Cleaning & Validation, MySQL, MongoDB, SQL Server
Deployment & MLOps	ONNX, Docker, FastAPI, Streamlit, AWS, CI/CD, FAISS, ChromaDB, Git/GitHub, Linux, Reproducible Experiment Design, Latency & Resource Profiling
Programming	Python, SQL, C++, C, C#, Go, JavaScript, MATLAB, Bash

SELECTED MACHINE LEARNING PROJECTS

- SimCLR-Vision-SSL — Self-Supervised Learning with Deployment-Ready Retrieval** 2026
PyTorch · SimCLR · SupCon · ONNX · FAISS
- Built a full self-supervised training pipeline and ran a **42-experiment augmentation sweep** to isolate which transformations actually drive representation quality.
 - Extended the model with a semi-supervised **SupCon** objective and reached **84.30% top-1 accuracy**, evaluated on both classification and image-retrieval tasks.
 - Exported the trained encoder to **ONNX** and served embeddings through a **FAISS** index, making the pipeline deployment-ready rather than notebook-only.
- SCRAP — Satellite Collision Risk Assessment & Prediction** 2026
Python · Scikit-learn · Supervised Learning · Feature Engineering
- Built a supervised ML pipeline that predicts satellite collision risk using only conjunction data available **48+ hours** before closest approach — deliberately constrained to match the real operational decision window rather than maximising offline accuracy.
 - Handled a heavily imbalanced target and engineered features from raw conjunction data, evaluating with metrics appropriate to rare, high-cost events.
- RL-Cloud-Autoscaler — Deep RL for Cloud Resource Provisioning** 2026
Python · PPO · DQN · Gymnasium
- Built a **custom Gymnasium environment** simulating cloud workload dynamics and trained **PPO** and **DQN** agents to optimise the **cost-latency trade-off**, benchmarked against static and threshold-based autoscaling baselines.
- Horus-OSINT — Fine-Tuned LLM Served on AWS** 2026
Meta-Llama-3-8B · Fine-Tuning · AWS · Python

- Fine-tuned **Meta-Llama-3-8B** on **159,826** GTD/GDELT records — dataset construction through training and evaluation — and deployed it on **AWS** as an interactive assistant.

Additional Projects

- **AudioShield** — CNN trained on mel-spectrograms for deepfake audio detection, served with real-time inference through a Streamlit application.
- **Verbosity-Aware Decoding** — MSc research on training-free token reduction at decoding time, benchmarked on MT-Bench, XSum and SQuAD-v2 with latency and GPU-energy profiling.
- **Human vs. Agentic Pull Requests** — reproducible GitHub-mining and analysis pipeline over large-scale open-source repositories.
- **Also:** Vitalism Solution (rPPG) · Diabetes Onset Prediction · ATM Transaction Analysis · Automated Multimodal Agent.

PROFESSIONAL EXPERIENCE

Machine Learning Instructor

National Telecommunication Institute (NTI)

Jul 2025 – Sep 2025
Cairo, Egypt

- Delivered a **90-hour** hands-on ML program for engineering and CS students covering supervised learning, PCA and neural networks, with practical lab sessions throughout.

AI in Cybersecurity Intern

e& Egypt — via NTI (On-the-Job Training)

Jun 2025 – Jul 2025
Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service)

National Company for Roads Building & Development

Feb 2024 – Mar 2025
Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**.
- Co-developed an internal data-management and reporting desktop application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**.

Online Instructor — AI, Algorithms & Data Science

Udemy & YouTube

2020 – Present
Remote

- Built and taught Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

AWS Academy — ML Foundations, ML for NLP, Data Engineering	Machine Learning Specialization — DeepLearning.AI
HCIP – AI (Huawei)	Microsoft Certified: Data Analyst Associate
ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)	AI in Cybersecurity — NTI (420 hrs)
1st Place — Smart Cities Hackathon 2022	Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.